# Paleomagnetism of Some Mesozoic Intrusives and Tuffs from Eastern Australia

W. A. ROBERTSON[1]

*Department of Geophysics*
*Australian National University, Canberra*

*Abstract.* Paleomagnetic studies of the Brisbane tuff and the Noosa Heads and Mount Dromedary intrusive complexes show that the stable components of natural remanent magnetization have rather uniform directions (irrespective of sign) with steep inclinations. In each case the time of formation of the rocks is known with precision, either from stratigraphic or from radioisotope evidence, to be Middle Triassic, Upper Jurassic, and late Lower Cretaceous, respectively. Field and laboratory studies suggest that the stable magnetization originated at the time of formation and was parallel to the earth's field. The observations therefore appear to provide a record of the earth's field at known times in the past. A reversal of magnetization in the Noosa Heads complex is related to a radioisotope age of 140 m.y. A detailed comparison of thermal and magnetic cleaning methods, in which rock samples from the Mount Dromedary intrusive complex were used, yields the result that the latter is to be preferred in this instance.

## BRISBANE TUFF

The Brisbane tuff, which is rhyolitic, occurs as subhorizontal lenses at several horizons in the Ipswich coal measures, which contain a Middle Triassic flora [*Jones and de Jersey*, 1947]. The tuff appears to have filled depressions in the old land surface and not to have been tilted subsequently by more than a few degrees; at Stafford quarry (site S, Table 1) a horizontal boundary between two flows may be seen. Samples have been obtained from six sites, which are building-stone quarries in and around the city of Brisbane. The mean location is (27.47°S, 153.03°E). Charred remains of trees overwhelmed by the tuff suggest extrusion at high temperatures. The directions in all specimens measured initially (NRM) and after partial demagnetization in an alternating magnetic field of 150 oe are plotted in Figure 1. The mean site directions and relevant statistics are listed in Table 1. As a test of stability, a single specimen from each site was subjected to alternating magnetic fields increased in 150-oe steps up to 600 oe; a field of 150 oe gave the smallest scatter (Figure 2), and all specimens are treated in this field. The scatter remains small in fields up to 450 oe. Figure 3 shows the decrease of the intensity of NRM with increase in alternating magnetic field in specimens from the most stable (S1) and least stable (P) sites. Curves for other specimens lie between these two. The directions after treatment in high alternating fields (Figure 2) are least scattered in the specimen from site S1 and most scattered in the specimen from site P, and this is correlated with slow and rapid falloff of intensity, respectively.

## NOOSA HEADS INTRUSIVE COMPLEX

The country rock at Noosa Heads, 110 km north of Brisbane, consists of undeformed Triassic and Jurassic sediments. These have been intruded by stocks, dikes, and a sill with a wide range of composition. *Houston* [1959] considers, on petrological evidence, that the intrusions are comagmatic and that they were probably all intruded within a short time interval [*Ridley*, 1960]. Potassium-argon age determinations [*Evernden and Richards*, 1962] have been made on biotite from a granitic dike at site C, giving 137 m.y., and hornblende from a porphyrite sill at site H, giving 142 m.y.; this difference is less than the experimental error. The intrusions are therefore assigned to the Upper Jurassic on *Kulp*'s [1961] time scale. The results from the eight sites are given in Table 2. At four sites the directions become random after partial demagnetization and were excluded from the analysis.

[1] Now with the Geophysics Division, Geological Survey of Canada, Booth St., Ottawa, Ont., Canada.

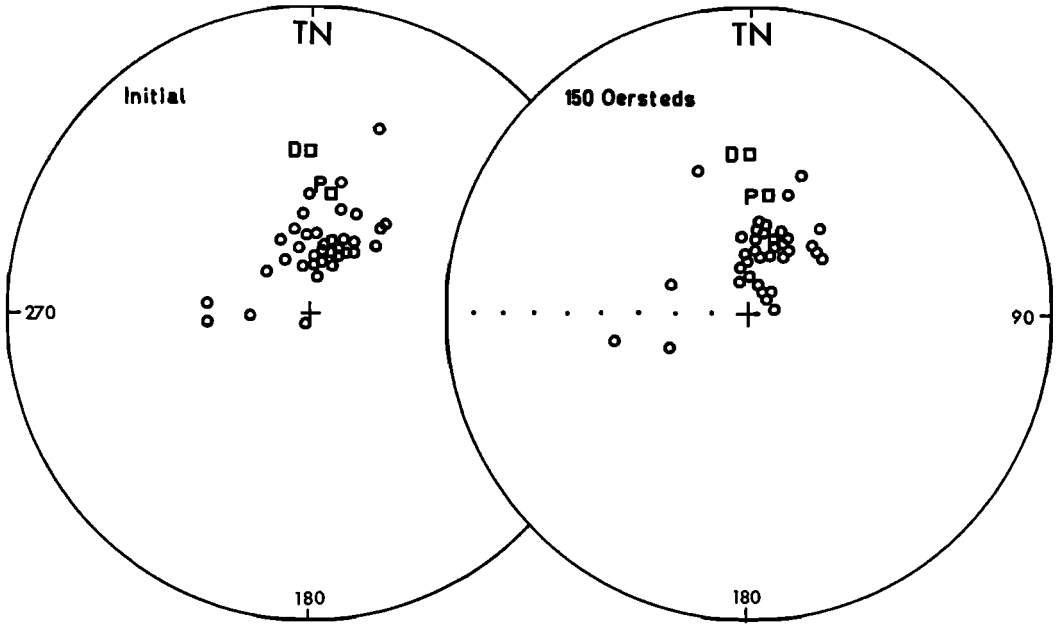



Fig. 1. Directions of natural remanent magnetization from the Brisbane tuff. North-seeking directions are plotted as circles on the upper hemisphere of an equal-area polar projection, before and after magnetic cleaning in an alternating field of 150 oe. The dipole ($D$) and present ($P$) field directions are shown.

The mean direction of the four remaining sites after cleaning is ($D = 51, I = -79, R = 3.94, \alpha_{95} = 13$). Specimens from sites H and B have a negative inclination, those from site D are of mixed polarity, and those from site F, which are considered to be the youngest, have a positive inclination. This reversal of direction suggests the possibility of a reversal of the earth's magnetic field at a well-defined time in the Jurassic 140 m.y. ago. One specimen from each site was demagnetized in incremental alternating magnetic fields, for which representative curves are shown in Figure 4. The intensity at the sites having initially random directions (C, A, G, and E) falls off rapidly, but at site F the intensity increases in low alternating fields, presumably owing to the removal of a component along the earth's present field.

## Mount Dromedary Intrusive Complex

*Geology.* The igneous rocks of the Mount Dromedary region, on the New South Wales coast 380 km by road south of Sydney, are intrusive into isoclinally folded Lower Paleozoic sediments. *Brown* [1930] tentatively concluded that the intrusion had a laccolithic form and that the variety of rock types was formed by differentiation in situ. A recent suggestion by *Joplin* [1962] that this is a ring-dike complex may modify this interpretation, but this is not immediately relevant to the present study. There is no evidence of tilting since emplacement.

The age of the intrusion was considered to be Permian on the grounds that the rocks were petrologically related to the lavas of known Permian age farther north on the Illawarra coast. More recently, K-Ar age determinations on biotites, hornblende and pyroxene from monzonite, shonkinite, and pyroxenite of the complex gave a small scatter at about 90 m.y. ago [*Evernden and Richards*, 1962], which places the intrusion at the top of the Cenomanian, in the upper part of the Lower Cretaceous period on *Kulp's* [1961] time scale. Samples were obtained from the major rock types and from the country rock heated by the intrusion. Sampling sites are shown on the geological sketch map (Figure 5).

*Sampling and results.* Oriented samples were collected from 28 sites. At 3 sites (B1, B2, and B6) the directions of NRM were random and remained random after partial demagnetiza-

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Brisbane Tuff Site Directions and Intensities

| Site | $S$ | $N$ | $M_n$ | | | | | $M_{150}$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $D$ | $I$ | $R$ | $a$ | $M_0$ | $D$ | $I$ | $R$ | $a$ | $M/M_0$ |
| W1 | 2 | 6 | 352 | −69 | 5.96 | 6 | 7.7 | 9 | −72 | 5.97 | 5 | 0.75 |
| S1 | 2 | 6 | 7 | −76 | 5.98 | 4 | 16 8 | 7 | −78 | 5.97 | 6 | 0.82 |
| S2 | 2 | 6 | 32 | −70 | 5.98 | 4 | 10.1 | 39 | −68 | 5.95 | 7 | 0.98 |
| K | 2 | 6 | 26 | −62 | 5.95 | 7 | 2.2 | 19 | −68 | 5.72 | 16 | 0.64 |
| P | 2 | 6 | 16 | −66 | 5.83 | 13 | 19.9 | 9 | −70 | 5.97 | 6 | 0.73 |
| U | 2 | 6 | 292 | −74 | 5 80 | 14 | 0.6 | 311 | −77 | 5.55 | 21 | 0.86 |
| Combined | 6 | 36 | 7 | −72 | 5.89 | 10 | | 11 | −74 | 5.94 | 7 | |

$M_n$ gives the NRM results, and $M_{150}$ gives those after partial demagnetization in an alternating magnetic field of 150 oersteds. $S$ is the number of samples and $N$ the number of specimens. $R$ is the length of the resultant, giving each specimen unit weight, and $D$ and $I$ specify its direction. $a$ is the half-angle of the cone of confidence at $P = 0.05$. The combined values have been obtained by applying Fisher's statistics to the mean site directions giving each unit weight. $M_0$ is the average intensity of magnetization in emu/cm$^3$ $\times$ $10^{-6}$. $M/M_0$ is the ratio of intensities after and before partial demagnetization. Grid references on the Brisbane 1-mile Military Sheet 183, Zone 8, are W1, 198948; S1 and S2, 191972; K, 200890; P, 194910; U, 172866.

tion. At sites B4 and G2 only one sample could be obtained because of collecting difficulties, and they are not considered further. The directions in the rocks at site E1 were reversed, and they showed curious magnetic properties that are described in a later section. Results from the remaining 22 sites are listed in Table 3, and specimen directions are plotted in Figure 6. The sites have been grouped into rock types using Brown's units supplemented by petrographic examination of thin sections from each site.

Because of the possibility of surface effects (incipient weathering, temperature hysteresis) changing the magnetization, it was thought desirable to obtain samples from holes blasted by gelignite. The effect of blasting was tested at sites P3, S2, and S3. Two samples, judged by eye to be unweathered, and later confirmed by examination of thin sections, were taken from the surface at these sites, and two samples were taken from the bottom of a hole blasted later at the same place. Comparison of the results is made in Table 4. At all three sites the directions of NRM are more consistent in samples taken from the bottom of blast holes than in surface samples from the same sites. At site P3 there is no significant difference in direction or intensity either before or after partial demagnetization in alternating magnetic fields between surface samples and samples from blast holes. At sites S2 and S3 initial directions from surface samples were scattered; after partial demagnetization in an alternating field of 150 oe, however, directions of these samples moved toward the direction of samples from the bottom of blast holes. It is concluded that the removal of the surface cap at rock exposures is in this case a desirable procedure and that the shock waves from blasting produce no deleterious effect on the NRM.

*Alternating magnetic field demagnetization of NRM.* From each of 11 sites representing all petrological groups four specimens were treated in alternating magnetic fields increasing up to 750 oe or until the scatter increased. Curves of precision (log $k$) and normalized intensity $M/M_0$ were plotted against alternating magnetic fields, and a representative selection of results is given in Figure 7. The estimate of precision, being based on only four specimens, is subject to considerable statistical fluctuations; the difference between the NRM precision for two sets from the same site (Table 3), which are later partially demagnetized, indicates the magnitude of these fluctuations. Mean directions for representative sites at the fields stated are shown in Figure 8.

The specimens from Mount Dromedary demagnetized in alternating magnetic fields fall into the following groups:

1. At sites B1, B2, and B6, initially random directions remain random throughout demag-

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

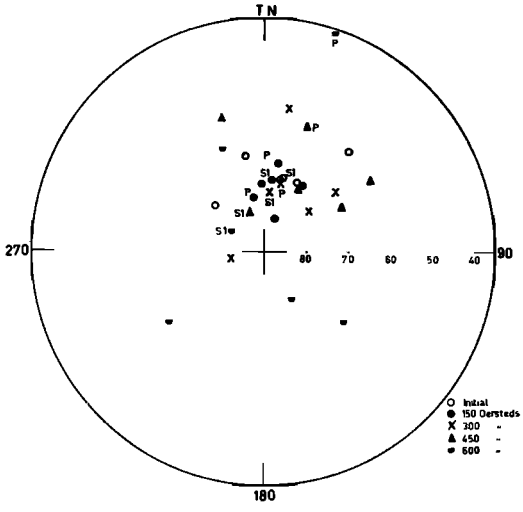



Fig. 2. Directions from partially demagnetized specimens of Brisbane tuff. One specimen from each collecting site was subjected to an alternating magnetic field whose peak value was increased by increments of 150 up to 600 oe. The directions are all north-seeking upward. Only the central part of the chart is shown.

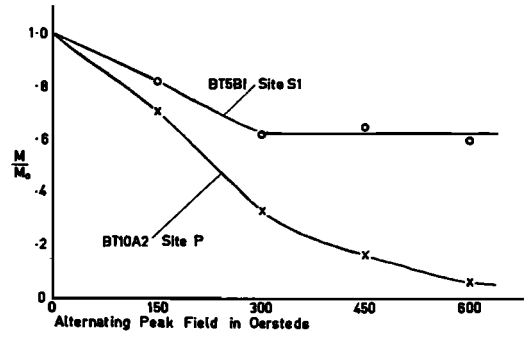



Fig. 3. Alternating magnetic field demagnetization of the Brisbane tuff. The ordinate gives the ratio of the intensity of magnetization remaining after treatment ($M$) in the alternating field whose peak value in oersteds is recorded along the abscissa, to the initial intensity of magnetization ($M_0$). The two extreme curves for sites S1 and P are shown; curves from specimens from the other sites fall between these two.

netization; the intensity falls off rapidly at first and then remains at a low fluctuating value.

2. At sites S1 and P2 widely scattered directions become tightly grouped after partial demagnetization and scatter in higher fields; the intensity falls off rapidly at first, and then more slowly. The scatter in fields greater than the coersivity arises from the 'scrambling' of the stable magnetization.

3. At sites O1, O2, O3, and S2 the directions are tightly grouped, and the precision remains nearly constant until it decreases steadily. Intensity in low fields for these sites falls slightly less rapidly than it does for group 2, but otherwise it is similar.

4. At sites M1 and V1 the directions are tightly grouped, and they remain so over a wide range of demagnetizing fields. The intensity of this group falls off slowly.

This grouping, which is gradational, may be explained by postulating the presence of a wide range of coercivities [*Graham*, 1953]. In group 1 stable components are apparently absent, in group 2 a large unstable component is readily removed by demagnetization, and in groups 3 and 4 the NRM is of a stable type, this being particularly marked in group 4. The intensity curves alone are only a broad criterion of stability, since rocks from group 2 may give a demagnetization curve indistinguishable from that of group 1, and dispersion of directions observed in sets of specimens from the same site must also be taken into account in this case.

*Thermal demagnetization of NRM.* Side by side with demagnetization in alternating magnetic fields, four different specimens from the same 11 sites were heated to successively higher temperatures and cooled in zero magnetic field. The maximum temperature in each run was maintained for about 5 minutes. Precision plotted against temperature (analogous to the alternating field curves) increased after heating at most sites and reached a maximum at a temperature about 100°C below the Curie point (Figure 7). The precision, however, changed slowly with increasing temperature followed by a rapid fall at the Curie point; at most sites there was a range of about 200°C in which the precision was not significantly different from the maximum.

Figure 9 shows a stereographic plot of the mean direction from four specimens at the temperatures shown for a stable (V1), partially stable (S1), and unstable (B6) site. The Curie point of NRM may be estimated from these thermal demagnetization curves in two different ways: (1) by selecting a point on the precision curve (Figure 7*c*) at which a random level is

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Noosa Heads Site Positions and Directions
(In probable order of intrusion.)

| Site | Description (oldest first) | $M_n$ S | N | D | I | R | a | $M_{af}$ $H_p\sim$ | N | D | I | R | a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| H | Hornblende porphyrite sill. Grid ref. 287205 | 2 | 6 | 219 | −20 | 4.80 | 38 | 300 | 6 | 267 | −85 | 5.59 | 20 |
| A | Andesitic dike. Grid ref. 287205 | 3 | 6 | 287 | −55 | 5.53 | 22 | 150 | 4 | Random | | | |
| G | Andesitic dike. Grid ref. 311205 | 2 | 6 | Random | | | — | 150 | 4 | Random | | | |
| B | Quartz diorite stock. Grid ref. 304219 | 2 | 6 | 33 | −66 | 5.86 | 12 | 150 | 6 | 43 | −73 | 5.94 | 7 |
| D | Quartz diorite stock. Foreshore at Coolum, 10 miles south of Noosa Heads. Grid ref. 283032 | 3 | 6 | 66 | −68* | 5.35 | 37 | 300 | 7 | 83 | −70* | 6.47 | 19 |
| C | Granitic dike. Grid ref. 302518 | 3 | 5 | 10 | −56 | 3.29 | 65 | 150 | 3 | Random | | | |
| E | Spilitic dike. Grid ref. 310203 | 2 | 6 | 357 | −67 | 4.93 | 35 | 150 | 4 | Random | | | |
| F | Spilitic dike. Grid ref. 300200 | 3 | 6 | 223 | +66 | 5.91 | 9 | 150 | 5 | 218 | +75 | 4.99 | 5 |
| | Combined from four used sites | | | | | | | | | 51 | −79 | 3.94 | 13 |

$M_n$ gives the NRM results, and $M_{af}$ gives the results after partial alternating field demagnetization in field $H_p\sim$ in oersteds. $S$ is the number of samples and $N$ the number of specimens. $R$ is the resultant giving each specimen unit weight, and $D$ and $I$ specify its direction. $a$ is the half-angle of the cone of confidence at $P = 0.05$. The number of specimens measured at random sites was the least necessary to test for randomness. The odd number of specimens at sites D and F was due to misfortunes during measurement.

* One reversed sample from site D was reversed in order to compute the resultant direction. Grid references are taken from Laguna Bay, 1-mile Military Sheet 143, Zone 8, except site D which is from Maroochydore Sheet 153, Zone 8.

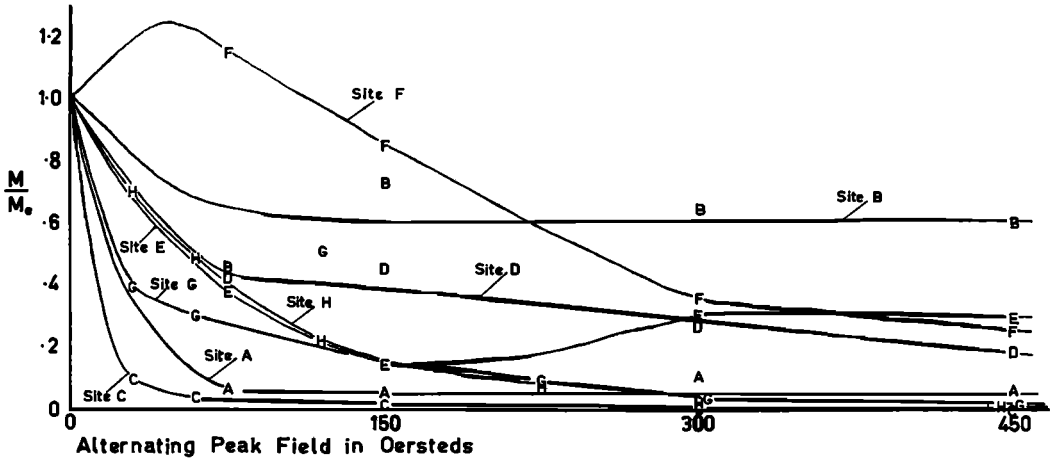



Fig. 4. Alternating magnetic field demagnetization of specimens from Noosa Heads. The coordinates are the same as for Figure 3. The letters indicate experimental points for the appropriate sites.

reached; (2) by taking as the lower limit the value of $T$ for which the gradient is greatest (Figure 7*d*), and as the upper limit the temperature at which the tangent at this point cuts the abscissa. The results obtained by both methods are given in Table 5 and show a broad measure of agreement. Low Curie point components of the NRM may be detected from inflections in the curves. Most of the plutonic rocks (sites A2, O1, G1, S1, and P1) had dominant Curie points

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

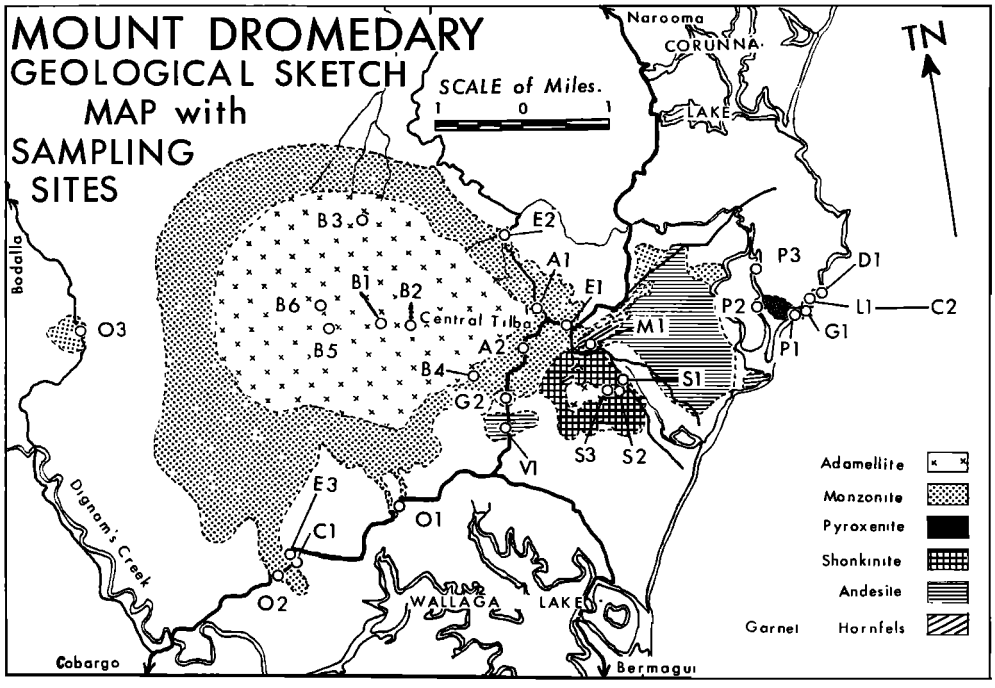



Fig. 5. Geological sketch map of Mount Dromedary showing sampling sites. (Modified after *Brown* [1930].)

between 540° and 560°C, whereas Curie points for the contact and dike rocks (sites L1, D1, C1, M1, and V1) ranged from 300°to 560°C. Only at site E, discussed later, was a lower Curie point evident.

*Thermal demagnetization of TRM.* One or more specimens from each site, 28 in all, were heated above 600°C and cooled in a steady field of 1.1 oe along the axis of the specimens and vertically upward. The acquired magnetization directions are plotted in Figure 10. No direction deviates by more than 5° from that of the applied field and most are within 2° of it; these deviations are probably due to experimental error. The *mean* direction is vertical ($I = -90$) with a precision $k = 1570$ ($N = 28$) and $\alpha_{95} = 2°$. Representative demagnetization curves of intensity for TRM and NRM are shown in Figure 7*d*, and the comparison of these with the demagnetization of NRM shows these features:

1. At sites A2, S1, and P2 the NRM tends to fall off more rapidly at low temperatures than the TRM, but at higher temperatures the curves are nearly parallel.

2. At sites O1, O2, and G1 the TRM and NRM curves are not significantly different.

3. The TRM and NRM curves are significantly different at sites V1 and M1. The TRM curves for these sites show a steady decrease of intensity from 350° to 550°C, whereas the NRM curve for site M1 shows a Curie point at about 350°C, and that for site V1 is at about 550°C.

The comparison suggests that specimens from group 1 may either contain low stability components that are removed by heating to 200°C or a chemical change may have occurred during heating in the range below 200°C. Specimens from group 2 have Curie points over 500°C and are not appreciably altered by heating. The magnetic mineral in group 3 appears to undergo partial chemical change during heating. The TRM curve at site V1, which is much smoother than that of NRM in the range 200° to 500°C, could be explained by mixing of formerly separated (exsolved) components forming a range of Curie points. The garnet-hornfels at site M1 are very inhomogeneous, so that it was possible to measure groups of garnet-rich and garnet-

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

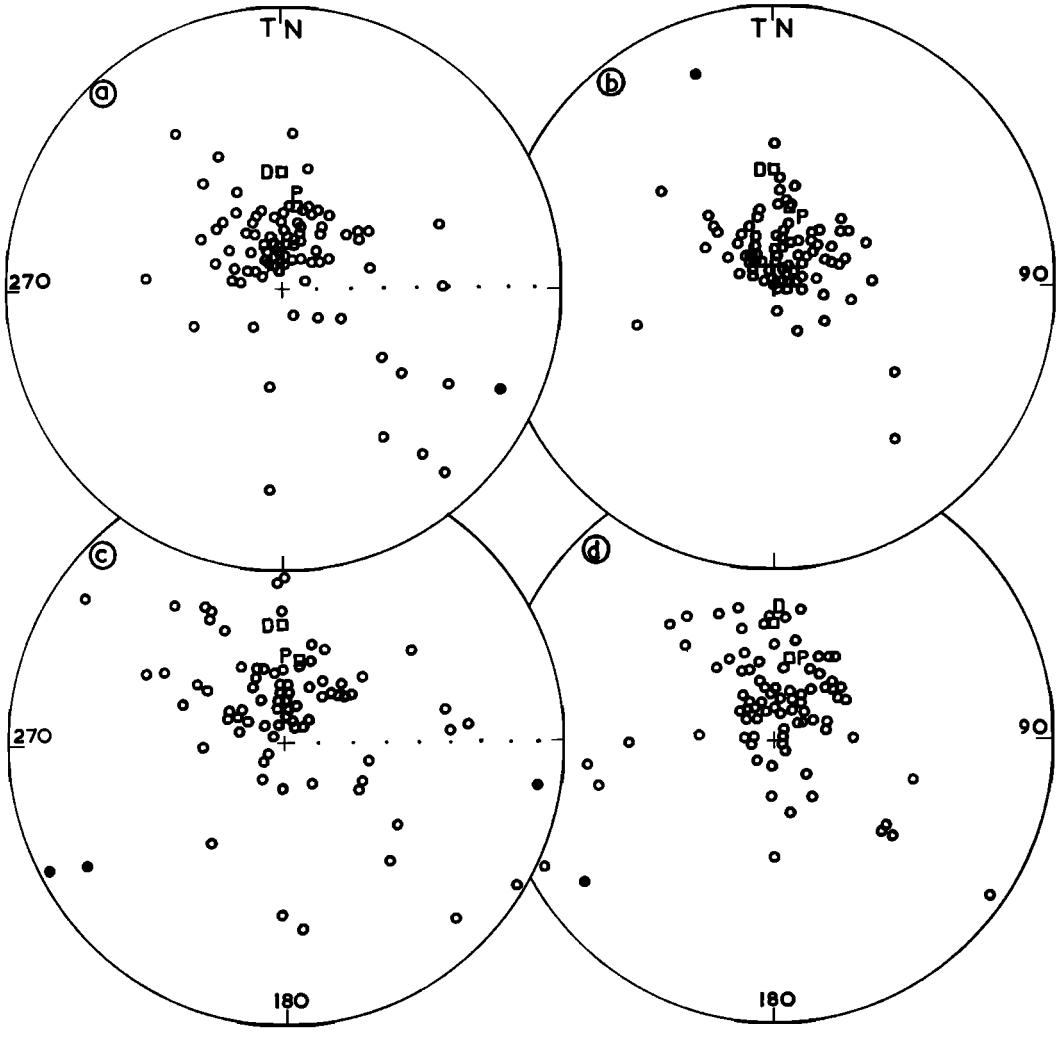



Fig. 6. Directions of natural remanent magnetization from Mount Dromedary. Conventions and projection as for Figure 1. Diagrams (*a*) and (*c*) show the initial directions for specimens that were later treated by alternating magnetic fields and heating, respectively, yielding the directions of diagrams (*b*) and (*d*). The present (*P*) and dipole (*D*) directions are also given.

poor specimens. The intensity of the garnet-rich specimens was greater by a factor of 10, suggesting that the magnetic mineral is associated with the garnet.

*Magnetic and thermal cleaning, a comparison of methods.* It has just been shown that the agreement between the directions observed in sets of specimens from selected sites increased to a maximum after treatment in alternating magnetic fields; the agreement remained approximately constant for most sites in the range of field strengths from 100 to 300 oe. Furthermore, partial thermal demagnetization gave maximum agreement in the temperature range 100° to 200°C below the Curie point. Similar partial demagnetization has been carried out on specimens from all the other collecting sites by heat and alternating fields in *each* case. Four specimens from the remaining sites were partially demagnetized in an alternating field of 150 oe, and

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

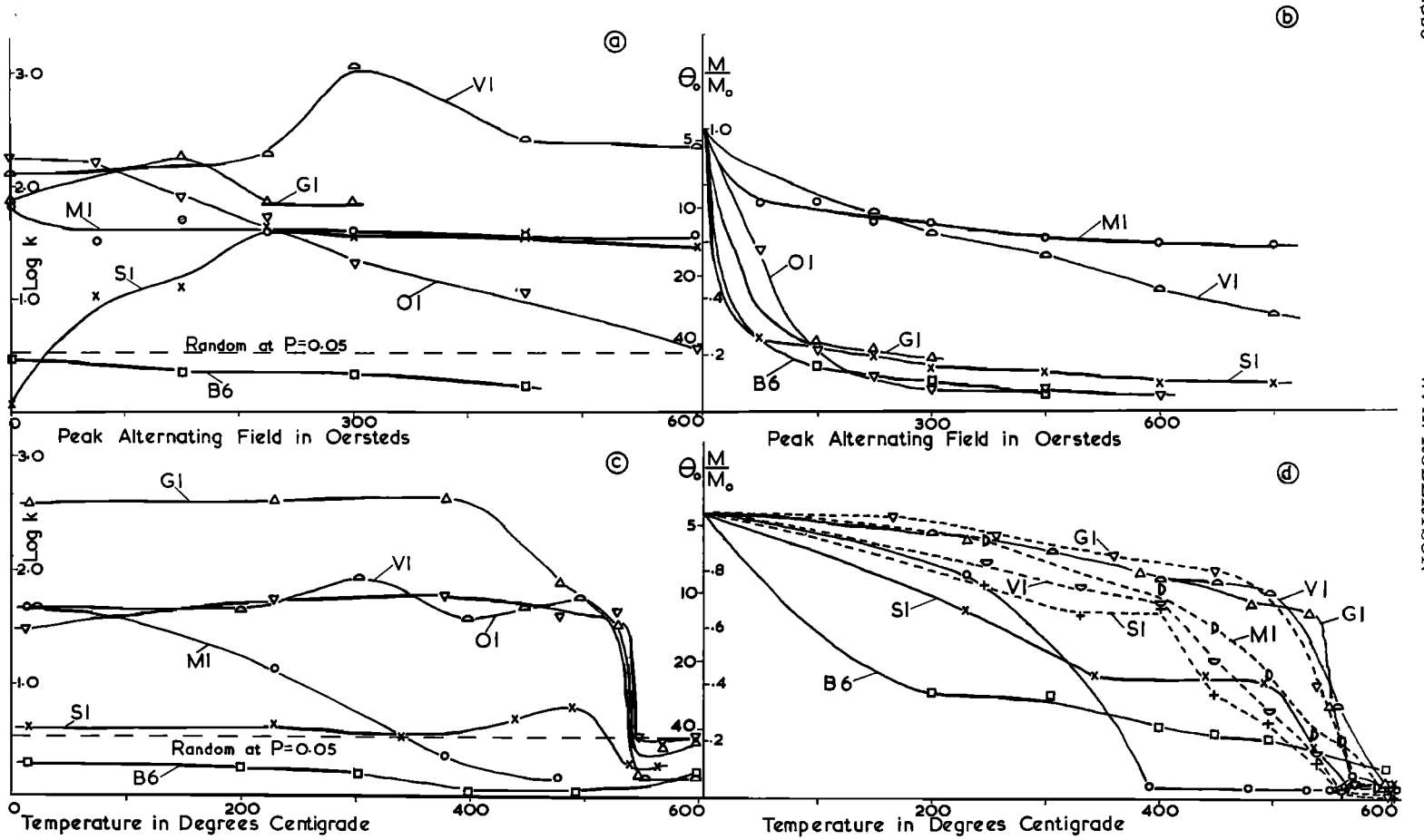



Fig. 7. Selected demagnetization curves from Mount Dromedary specimens. In (*a*) and (*c*) $\Theta_0$ is the circular standard deviation of the observations. The continuous lines are for demagnetization of NRM. In (*d*) the demagnetization of TRM is indicated by dotted lines. The numbers refer to sampling sites in Table 3.

2156202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Mount Dromedary Site Directions and Intensities

| Rock Type | Site | $M_n$ S | N | D | I | R | $M_0$ | Treatment: Alternating Field, oe | Treatment: Demag. Temp., °C | $M_{TR}$ D | I | R | $M/M_0$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Adamellite (banatite) | B3 | 3 | 4 | 240 | −81 | 3.83 | .44 | 300 | | 89 | −85 | 3.82 | .09 |
| | | 3 | 4 | 304 | −80 | 3.85 | .31 | | 465 | 282 | −59 | 3.57 | .32 |
| | B5 | 3 | 4 | 109 | −45 | 3.78 | .23 | 150 | | 118 | −62 | 3.54 | .17 |
| | | 3 | 4 | 106 | −44 | 3.43 | .17 | | 450 | 126 | −39 | 3.58 | .38 |
| Coarse monzonite | A1 | 3 | 4 | 13 | −70 | 3.96 | 1.6 | 225 | | 12 | −69 | 3.95 | .27 |
| | | 2 | 2* | 142 | −53 | 1.93 | 1.1 | | 380 | 134 | −70 | 3.68 | .38 |
| | A2 | 2 | 4 | 330 | −80 | 3.81 | .85 | 225 | | 27 | −78 | 3.95 | .13 |
| | | 2 | 4 | 320 | −67 | 3.94 | .58 | | 305 | 340 | −68 | 3.89 | .66 |
| Even monzonite | E2 | 3 | 4 | 330 | −47 | 3.94 | .20 | 225 | | 35 | −64 | 2.66 | .29 |
| | | 3 | 4 | 328 | −43 | 3.99 | .24 | | 201 | 326 | −50 | 3.93 | .81 |
| | E3 | 2 | 4 | 351 | −69 | 3.96 | 2.7 | 225 | | 338 | −74 | 3.95 | .18 |
| | | 2 | 4 | 347 | −69 | 3.96 | 2.8 | | 256 | 339 | −77 | 3.97 | .74 |
| Olivine monzonite | O1 | 2 | 4 | 9 | −74 | 3.98 | 2.8 | 75 | | 23 | −79 | 3.98 | .57 |
| | | 2 | 4 | 25 | −72 | 3.93 | 2.9 | | 380 | 33 | −80 | 3.95 | .82 |
| | O2 | 2 | 4 | 342 | −72 | 3.81 | 2.3 | 150 | | 348 | −78 | 3.89 | .23 |
| | | 2 | 4 | 341 | −62 | 3.77 | 4.2 | | 256 | 351 | −78 | 3.91 | .48 |
| | O3 | 2 | 4 | 13 | −70 | 3.98 | 5.1 | 225 | | 77 | −82 | 3.99 | .19 |
| | | 1 | 2 | 33 | −69 | 1.99 | 5.6 | | 450 | 26 | −83 | 1.99 | 1.03 |
| Gabbro | G1 | 3 | 4 | 24 | −82 | 3.96 | 5.5 | 150 | | 3 | −77 | 3.98 | .24 |
| | | 2 | 4 | 22 | −84 | 3.99 | 7.2 | | 380 | 42 | −80 | 3.99 | .80 |
| Pyroxenite | P1 | 3 | 4 | 95 | −87 | 3.50 | 6.2 | 225 | | 338 | −74 | 3.92 | .22 |
| | | 3 | 4 | 89 | −55 | 3.43 | 12.0 | | 340 | 33 | −61 | 3.93 | .42 |
| | P2 | 2 | 4 | 229 | −83 | 3.83 | 2.3 | 75 | | 8 | −84 | 3.97 | .40 |
| | | 2 | 4 | 272 | −76 | 3.67 | 1.7 | | 256 | 104 | −85 | 3.90 | .78 |
| | P3 | 4 | 8 | 32 | −71 | 7.90 | 2.4 | 300 | | 30 | −74 | 6.97 | .27 |
| | | 4 | 4 | 345 | −66 | 3.37 | 2.6 | | 450 | 40 | −75 | 3.96 | .67 |
| Shonkinite | S1 | 3 | 4 | 66 | −25 | 1.57 | .74 | 225 | | 2 | −75 | 3.93 | .20 |
| | | 3 | 4 | 146 | −23 | 3.25 | .96 | | 490 | 122 | −60 | 3.52 | .42 |
| | S2 | 4 | 8 | 252 | −83 | 6.59 | 2.2 | 150 | | 349 | −83 | 7.74 | .11 |
| | | 4 | 4 | 248 | −59 | 2.99 | 2.2 | | 450 | 245 | −74 | 3.24 | .49 |
| | S3 | 2 | 4 | 56 | −62 | 3.99 | .46 | 450 | | 65 | −70 | 3.99 | .18 |
| | | 4 | 4 | 303 | −69 | 2.38 | .46 | | 450 | 284 | −78 | 3.44 | .32 |
| Lamprophyre | L1 | 2 | 2 | 333 | −76 | 1.98 | 2.6 | 225 | | 22 | −68 | 1.84 | .09 |
| | | 2 | 4 | 353 | −79 | 3.93 | 2.1 | | 230 | 22 | −79 | 3.92 | .67 |
| Porphyry | D1 | 2 | 4 | 316 | −66 | 3.75 | .06 | 75 | | 308 | −66 | 3.55 | .24 |
| | | 2 | 4 | 354 | −57 | 3.89 | .04 | | 201 | 356 | −60 | 3.97 | .62 |
| Baked contacts | C1 | 2 | 4 | 350 | −80 | 3.99 | 3.0 | 225 | | 339 | −79 | 3.99 | .39 |
| | | 2 | 4 | 357 | −78 | 3.99 | 4.3 | | 340 | 353 | −78 | 3.99 | .63 |
| | C2 | 2 | 4 | 65 | −85 | 3.62 | .49 | 225 | | 50 | −80 | 3.88 | .29 |
| | | 2 | 4 | 285 | −81 | 3.94 | .65 | | 256 | 333 | −74 | 3.92 | .43 |
| Garnet hornfels | M1 | 2 | 4 | 342 | −78 | 3.95 | .04 | 150 | | 297 | −84 | 3.94 | .67 |
| | | 2 | 4 | 313 | −82 | 3.94 | .05 | | 230 | 355 | −88 | 3.77 | .78 |
| Andesite | V1 | 2 | 4 | 43 | −72 | 3.98 | 3.2 | 300 | | 46 | −72 | 3.99 | .63 |
| | | 3 | 4 | 47 | −73 | 3.90 | 2.4 | | 305 | 46 | −72 | 3.97 | .87 |

$M_n$, $S$, $N$, $D$, $I$, $R$, and $M/M_0$ have the same meaning as in Table 1. $M_0$ is the average intensity in emu/cm$^3$ $\times$ $10^{-3}$. $M_{TR}$ specifies results after partial demagnetization.

* The NRM from only one specimen from each sample was measured; the results after heating are from four specimens, two from each sample.

a separate set of four specimens from these sites was heated to 160°C, and those from plutonic rocks reheated to 450°C.

In Table 3 results are given from all sets of specimens before treatment, and after treatment both in the alternating field and at the temperature that yielded the highest precision. At sites such as E3, G1, C1, and O1, there is no signifi-

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. A Test on the Effect of Blasting

| Site | $H_p\sim$ | $S$ | $N$ | Surface Specimens | | | | Specimens from Blast Holes | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $D$ | $I$ | $R$ | $M$ | $D$ | $I$ | $R$ | $M$ |
| P3 | 0 | 2 | 4 | 26 | −71 | 3.98 | 2.2 | 42 | −72 | 3.93 | 2.6 |
| P3 | 150 | 2 | 4 | 37 | −72 | 3.91 | 1.0 | 35 | −75 | 3.99 | 1.1 |
| S2 | 0 | 2 | 4 | 229 | −46 | 2.39 | 2.8 | 17 | −76 | 3.98 | 1.5 |
| S2 | 150 | 2 | 4 | 348 | −81 | 3.78 | 0.28 | 352 | −85 | 3.97 | 0.22 |
| S3 | 0 | 2 | 4 | 269 | −36 | 3.23 | 0.37 | 56 | −62 | 3.99 | 0.54 |
| S3 | 150 | 2 | 4 | 275 | −43 | 2.93 | 0.16 | 83 | −70 | 3.96 | 0.26 |

$H_p\sim$ is the alternating magnetic field in which the specimens were partially demagnetized. $S$ is the number of specimens and $N$ the number of samples. $R$ is the length of the resultant giving each specimen unit weight and $D$ and $I$ specify its direction. $M$ is the average intensity of magnetization for the group in emu/cm$^3$ × $10^{-3}$.

cant difference in direction between groups of specimens from the same site, either before or after partial demagnetization by either method. The small dispersions observed are probably due to experimental errors. At most other sites both methods decreased the scatter; at sites S1, S2, and C1 alternating fields produced the greater improvement whereas at site O2 heating was more efficient. Both methods increased the scatter at sites B3 and E2, whereas alternating fields increased the scatter at sites B5, L1, and D1, and heating increased scatter at sites A2 and M1. Considered site by site this comparison yields a picture of much variability.

A comparison between results from different sites is made in Table 6. The sites are divided into three geological groups. Partial demagnetization in alternating magnetic fields gives more consistent results than thermal partial demagnetization for the plutonic intrusions, whereas

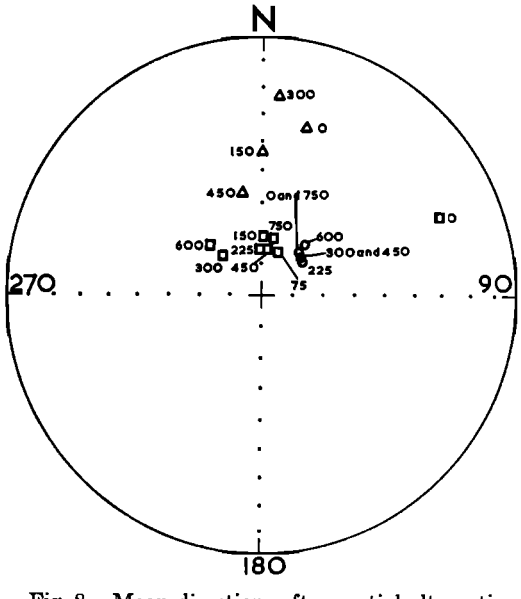



Fig. 8. Mean directions after partial alternating field demagnetization at some Mount Dromedary sites. The numbers show the peak value of the alternating magnetic field in oersteds applied to the specimens. Circles represent site V1, squares represent site S1, and triangles represent site B6.

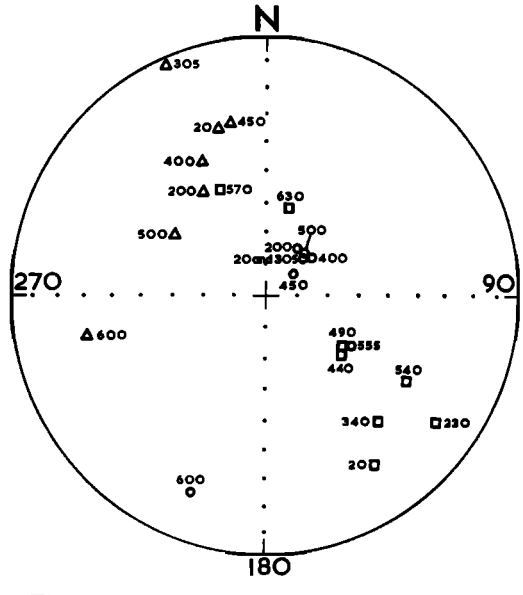



Fig. 9. Mean directions after partial thermal demagnetization at some Mount Dromedary sites. The numbers indicate the temperatures (°C) to which the specimens were raised. Circles represent site V1, squares represent site S1, and triangles represent site B6.

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

for the dike and contact rocks the reverse is true. There is no significant difference between the means for these groups, and this indicates that there is no important secondary component in any one group after cleaning. The over-all means after magnetic or thermal cleaning are not significantly different. The mean is more accurately defined after magnetic cleaning than after thermal cleaning. The greater over-all precision after magnetic cleaning is evident in Figure 6.

*Mineragraphic investigations.* Results of polished sections and powder photography work are summarized in Table 7. The polished sections examined may not be representative of the rock as a whole, since the minerals in some sam-

TABLE 5. Estimation of Curie Points of NRM

| Site | From Precision Curve, °C | From Intensity Curve, °C |
|---|---|---|
| Plutonic rocks | | |
| B3 | Indeterminate | Indeterminate |
| A2 | 560 | 540–560 |
| O1 | 550 | 560–590 |
| G1 | 550 | 550–575 |
| P1 | 540 | 530–580 |
| S1 | 520 | 530–570 |
| Dikes | | |
| L1 | 320 | 275–380 |
| D1 | 330 | 250–340 |
| Contact rocks | | |
| C1 | 555 | 525–555 |
| M1 | ~345 | 300–400 |
| V1 | 545 | 550–590 |

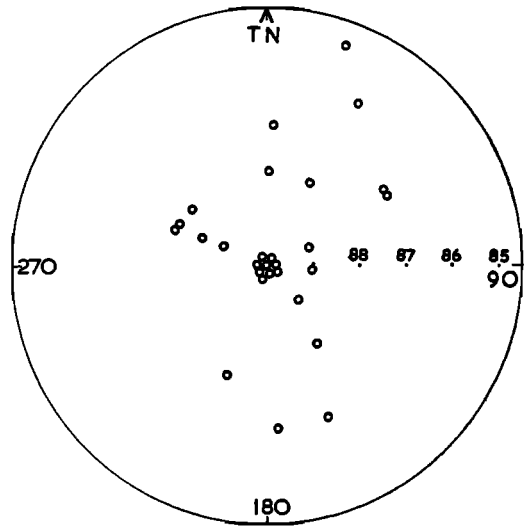



Fig. 10. TRM directions. The directions from 28 specimens heated to 600°C and cooled in a field directed along the axis of the specimens. The center only of the net is shown.

TABLE 6. Comparison of Groups at Mount Dromedary

| Group | Number in Group | Partial Alternating-Field Demagnetization | | | | | | | | | Partial Thermal Demagnetization | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Initial | | | | Treated | | | | | Initial | | | | Treated | | | | |
| | | *D* | *I* | *R* | *a* | *D* | *I* | *R* | *a* | *Q* | *D* | *I* | *R* | *a* | *D* | *I* | *R* | *a* | *Q* |
| 1 | 10 | 3 | −76 | 9.38 | 13 | 31 | −80 | 9.75 | 8 | 2.4 | 8 | −79 | 8 93 | 18 | 355 | −83 | 9.10 | 16 | 1 2 |
| 2 | 7 | 55 | −76 | 6.42 | 20 | 15 | −78 | 6.93 | 7 | 8.3 | 179 | −88 | 5.88 | 29 | 59 | −84 | 6.67 | 15 | 3 4 |
| 3 | 4 | 333 | −78 | 3.95 | 12 | 349 | −79 | 3.90 | 17 | 0.5 | 339 | −76 | 3.91 | 16 | 355 | −76 | 3.93 | 15 | 1.2 |
| Average of Groups | 4* | 23 | −78 | 3.96 | 10 | 24 | −78 | 3.98 | 7 | 2.0 | 15 | −81 | 3.95 | 12 | 24 | −80 | 3.97 | 9 | 1.5 |
| Average of Sites | 22* | 17 | −78 | 20.60 | 8 | 19 | −79 | 21.53 | 5 | 3.0 | 3 | −83 | 19.56 | 11 | 13 | −82 | 20.61 | 8 | 1.8 |

Group 1 is from Mount Dromedary, group 2 is from Little Dromedary, and group 3 is comprised of dike and contact rocks. *D*, *I*, *R*, and *a* mean the same as in Table 1. *Q* is the ratio of the precision after partial demagnetization to that before.

* Site V1, not included in groups 1 to 3, as it is probably a lava flow, is included in the over-all mean statistic.

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 7. Opaque Minerals

| Site | Rock | Percentage of 'Opaques' in Section | Opaque Minerals Identified in Polished Section in Order of Abundance | X-Ray Powder Photograph |
|---|---|---|---|---|
| B1 | Adamellite | <2 | Magnetite | |
| B3 | Adamellite | <1 | Mineral resembling ilmenite | |
| A1 | Coarse monzonite | 3 | Magnetite-ilmenite intergrowths, magnetite, ilmenite, pyrite, chalcopyrite | Magnetic fraction yielded a magnetite diffraction pattern only, with a lattice parameter of 8.39 A. |
| E1 | Even-grained monzonite (reversely magnetized) | 2 | Pyrrhotite, ilmenite, magnetite, magnetite-ilmenite intergrowths, pyrite, chalcopyrite, and marcasite | A crystal, selected to check identification, yielded a pyrrhotite diffraction pattern. |
| E3 | Even-grained monzonite | 3 | Magnetite-ilmenite intergrowths, magnetite, ilmenite | |
| O1 | Olivine monzonite | 4 | Ilmenite, magnetite, ilmenite-magnetite intergrowths, chalcopyrite | |
| G1 | Gabbro | 6 | Ilmenite, pyrite, pyrrhotite, magnetite | |
| P1 | Pyroxenite | 10 | Magnetite, ilmenite, chalcopyrite, pyrrhotite, marcasite | Magnetic fraction yielded only a magnetite diffraction pattern, with a lattice spacing of 8.39 A. |
| S1 | Shonkinite | 3 | Magnetite-ilmenite intergrowths, magnetite, ilmenite, chalcopyrite, pyrite, pyrrhotite, and marcasite | Magnetic fraction yielded only a magnetite diffraction pattern, with a lattice spacing of 8.39 A. |
| C1 | Schist baked by monzonite | 8 | Ilmenite, chalcopyrite | |
| C2 | Schist baked by lamprophyre | <2 | Pyrrhotite | |
| M1 | Garnet hornfels | <1 | Pyrite | |
| V1 | Andesite | 5 | Magnetite-ilmenite intergrowths, magnetite, ilmenite, pyrite | |

ples, such as those from sites G1, P1, O1, and M1, were unevenly distributed. It was not always possible to identify all the opaque minerals present, particularly if they occurred in extremely small grains, as at sites M1 and C1. Minerals of the iron-titanium oxide series predominate. Pyrrhotite is common, and nonmagnetic pyrite and chalcopyrite are also present. The high Curie points and the small lattice parameters (such as those at sites A1, P1, and S1) suggest that the magnetite present is nearly pure, with the titanium almost completely exsolved as ilmenite.

*The monzonite at site E1, a possible self-reversal?* The magnetization directions of specimens from site E1, which are from an even-grained monzonite, initially gave a tight group in the opposite sense to those from other sites (Figure 11). It is possible that the earth's field was reversed at the time of cooling, but this seems at first sight unlikely because no other reversed rocks were found in the complex. The rock has magnetic properties that are different from any other studied from the complex.

Four specimens heated to 230°C maintained high precision in the reversed direction, but they became widely scattered after heating to 340°C, and random at 440°C. The intensity of magneti-

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

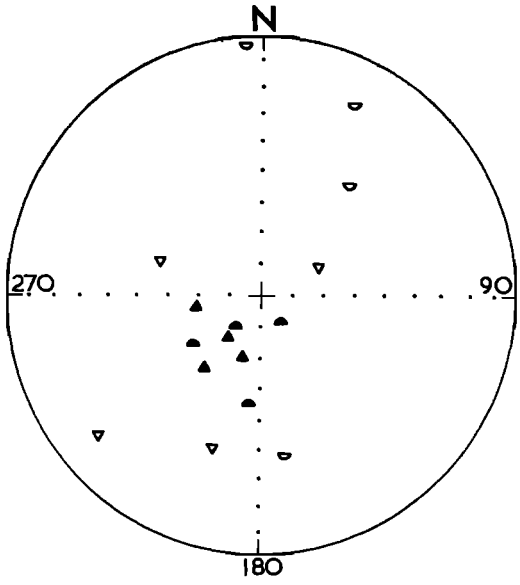



Fig. 11. Directions from site E1. Solid triangles are the initial directions from specimens later partially demagnetized by alternating magnetic fields; open triangles are the direction of the same specimens after treatment in fields of 300 oe. Solid half-circles are the initial directions from specimens later partially thermally demagnetized; open half-circles are the directions of the same specimens after heating to 540°C.

zation decreased rapidly to 340°C and then decreased slowly to 565°C. The intensity of specimens given a TRM fell off less rapidly with temperature. The trend of directions was opposed to the initial direction at 540°C, although the scatter of points was very high. Demagnetization by alternating magnetic fields gave a similar but more clear-cut result. Their directions, initially well grouped, became random at 225 oe and were widely scattered with negative inclinations after treatment at 300 oe (Figure 11), and at 450 oe the scatter increased. The intensity fell to about 3 per cent of the initial intensity after partial demagnetization at 225 oe, and in higher fields it decreased slowly. It would appear that, after heating above 340°C and partial demagnetization in an alternating magnetic field of 225 oe, the directions are opposed to the NRM direction, but the scatter is very high. Examination of two polished sections revealed that the dominant magnetic mineral was pyrrhotite with a grain size of 0.5 mm, and magnetite about a tenth of this size was associated with it.

TABLE 8. Mean Rock Unit Directions

| Rock Unit | Treatment | *B* | *D* | *I* | *R* | *a* |
|---|---|---|---|---|---|---|
| Mount Dromedary | A.F. 75–450 | 22 | 19 | −79 | 21.53 | 5 |
| Mount Dromedary | Th. 201–440 | 22 | 13 | −82 | 20.61 | 8 |
| Noosa Heads complex | A.F. 150–300 | 4 | 51 | −79 | 3.94 | 13 |
| Brisbane tuff | A.F. 150 | 6 | 11 | −74 | 5.94 | 7 |

*B* is the number of sites. *D*, *I*, *R*, and *a* have the same meaning as in Table 1. A.F. is partial demagnetization in an alternating magnetic field of peak value stated in oersteds. Th. is partial demagnetization by heating to the temperature stated in degrees centigrade.

It is interesting to speculate and suggest that the reversed NRM arose from some interaction between the pyrrhotite and magnetite. TRM imposed in the laboratory was parallel to, and in the same sense as, the applied field. Possible self-reversal properties have also been noticed in other rocks containing pyrrhotite [*Almond et al.*, 1956; *Everitt*, 1962]. Another possible but unlikely alternative is that the earth's magnetic field reversed in the time at which the rock cooled between the Curie points of magnetite and pyrrhotite. The present study is inconclusive, and the results are given to complete the field account of this work. The question whether this is a self-reversal, or whether it reflects a true reversal of the earth's field, is left open.

*Discussion.* The following points indicate that the direction obtained at Mount Dromedary (except at site E1) is that of the field at the time of cooling:

1. Studies of alternating field and thermal demagnetization show that components of high stability have highly consistent directions of magnetization. This consistency is found between rocks formed under different physical and chemical conditions. The samples studied are from dikes and from the margins and interiors of over- and undersaturated plutonic masses, in

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

which stresses, load, and chemical equilibria conditions are likely to have been variable.

2. Contact rocks heated by the intrusions are magnetized in the same direction as the intrusions, indicating that the magnetization is thermoremanent, acquired at the time of intrusion.

3. At some sites demagnetization curves of TRM and NRM are identical, indicating that the NRM is of a TRM type.

4. One or more specimens from all sites cooled in the laboratory from 600°C in a known magnetic field acquired magnetization accurately parallel to that of the applied field, suggesting that the specimens are magnetically isotropic.

## Conclusions

Table 8 gives the mean directions of the stable magnetization and associated confidence limits of the three rock groups discussed in this paper. The ages of these rock groups are accurately known, and the stability of the directions has been tested; therefore the results may be regarded as reasonable estimates of the direction of the earth's field at these times of formation.

## References

Almond, Mary, J. A. Clegg, and J. C. Jaeger, Remanent magnetism of some dolerites, basalts and volcanic tuffs from Tasmania, *Phil. Mag., 1,* 771–782, 1956.

Brown, Ida A., The geology of the south coast of New South Wales, 3, The monzonitic complex of the Mount Dromedary district, *Proc. Linnean Soc. N. S. Wales, 55,* 637–698, 1930.

Everitt, C. W. F., Self-reversal of magnetization in a shale containing pyrrhotite, *Phil. Mag., 7,* 831–843, 1962.

Evernden, J. F., and J. R. Richards, Potassium-argon ages in eastern Australia, *J. Geol. Soc. Australia, 9,* 1–49, 1962.

Graham, J. W., Changes of ferromagnetic minerals and their bearing on magnetic properties of rocks, *J. Geophys. Res., 58,* 243–260, 1953.

Houston, B., Honours thesis, Brisbane, 1959.

Jones, O. A., and N. J. de Jersey, The flora of the Ipswich coal measures—morphology and floral succession, *Papers Dept. Geol. Univ. Queensland, 3,* 1–88, 1947.

Joplin, Germaine A., An apparent magmatic cycle in the Tasman geosyncline, *J. Geol. Soc. Australia, 9,* 51–69, 1962.

Kulp, J. L., Geologic time scale, *Science, 133,* 1105–1114, 1961.

Ridley, W. F., The Maryborough basin, *J. Geol. Soc. Australia, 7,* 197–301, 1960.

(Manuscript received October 11, 1962; revised January 11, 1963.)

21562202, 1963, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ068i008p02299 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License